

1(a)(i)	<i>independent:</i> sex / male or female / boy or girl ; <i>dependent:</i> reaction time ;	2						
1(a)(ii)	categoric ;	1						
1(a)(iii)	<i>any 6 of:</i> 1. <i>ref. to</i> suitable number of people in total ; 2. same / similar ages ; 3. same number boys and girls in each group ; 4. <i>ref. to</i> consent form (e.g. using data / children / AW) ; 5. <i>ref. to</i> standard conditions for the test, e.g. same (type of) computer / tablet / software / same colour change ; 6. <i>ref. to</i> same, stated, room conditions ; 7. <i>ref. to</i> avoiding people with conditions affecting reaction time e.g. poor eyesight / colour-blind / neurological disorders ; 8. no, (named) drugs / medications / caffeine / alcohol ; 9. <i>ref. to</i> testing at same time of day / stated time ; 10. <i>ref. to</i> using same hand to push the button ; 11. minimum of three tests on each subject and taking a mean OR minimum of three subjects of each sex and taking a mean ; 12. low / medium risk experiment / parent or guardian present for children / carer for vulnerable adults ;	6						
1(a)(iv)	<i>idea of</i> processing delay from computer / AW ;	1						
1(b)	<table border="1"> <tr> <td>mode</td><td>0.26</td><td>0.25</td></tr> <tr> <td>median</td><td>0.25 ;</td><td>0.26</td></tr> </table>	mode	0.26	0.25	median	0.25 ;	0.26	1
mode	0.26	0.25						
median	0.25 ;	0.26						

Question	Answer	Marks
1(c)(i)	comparing (two) <u>means</u> / continuous data / normal distribution ;	1
1(c)(ii)	there is no (significant) difference in the (mean) reaction <u>time</u> (in right handed people) when using the left or right hand ;	1
1(c)(iii)	<u>20</u> ;	1
1(c)(iv)	<u>less than</u> 2.09 / less than the critical value (at $p = 0.05$) ;	1
1(d)(i)	7.9 (trial 11, student 2, before measurement) ;	1
1(d)(ii)	<i>idea that</i> affected data should be left out ;	1
1(d)(iii)	student 3 (no mark) (as their) standard deviations / s, are largest ;	1
1(d)(iv)	<i>supports</i> <i>any 1 of:</i> 1. <i>idea that</i> mean measurement for students was close(r) to, the actual measurement / 7.5, after feedback had been given ; 2. mean +/- s before feedback does not overlap with mean +/- s after feedback for, students 2 / 3 ; 3. AVP ; <i>does not support</i> <i>any 1 of:</i> 4. small sample size / only 3 students tested ; 5. mean +/- s for student 1 before and after feedback overlap 6. <i>idea that</i> improvement in accuracy in student 3 is small / 1.5 to 1.4 (cm) change in closeness to mean ; 7. <i>idea of</i> no statistical analysis to determine if difference in results is significant or not ; 8. AVP ;	2